

eDCU Life Time Tester

Operation User Manual

Document ID: 23-10-051-VTI-CCAIC-UM

Rev.: 2.0

Safran Data Systems Pvt Ltd

	Name	Signature	Date
Prepared By	Ananth A		
Reviewed By	Prabhakar T J		
Approval By	Midhunseelan O S		

Customer

	Name	Signature	Date
Reviewed By			
Approved By			

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 1

Document ID	23-10-051-VTI-CCAIC-UM
Document Name:	Operation User Manual
Document No.:	R/PP-03/25
Project/Product Name:	eDCU Life Time Tester
Project/Product Code:	23-10-051-VTI-CCAIC
Revision Status:	2.0

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 2

Change Summary

Release No.	Date	Changed By	Change Description
Version 1.0	18-11-2024	Ananth A	Initial Version

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 3

Table of Contents

Change Summary.....	2
1.1 Desktop Panel.....	4
1.2 Main Screen::	5
1.3 Main Login:	6
1.4 Main Login Exit.....	7
1.5 Auto Mode:	8
1.6 Test stand Login:.....	9
1.7 Sequence Selection:.....	10
1.8 Test Windows:.....	11
1.9 Socket Entry Windows:	12
1.10 UUT Info Entry Windows:	13
1.11 UUT Info Entry Scroll Windows:	14
1.12 Test Selection Windows:	15
1.13 Schema File Location:	16
1.14 Schema Selection Windows:	17
1.15 Start the Chamber and Test Start Windows:.....	18
1.16 Running Windows of Monitor:.....	19
1.17 Terminate Test Procedure:	20
1.18 Error Pop-UP Test Procedure:	21
1.19 Exit Application:	22

Project Name: eDCU Life Time Tester	Document ID: 21-09-041-PHL-CBAIC-UM	Version:1.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: User Manual		Page 4

1.1 Desktop Panel

By clicking the “Application” button in the main screen, below shown User window will appear.

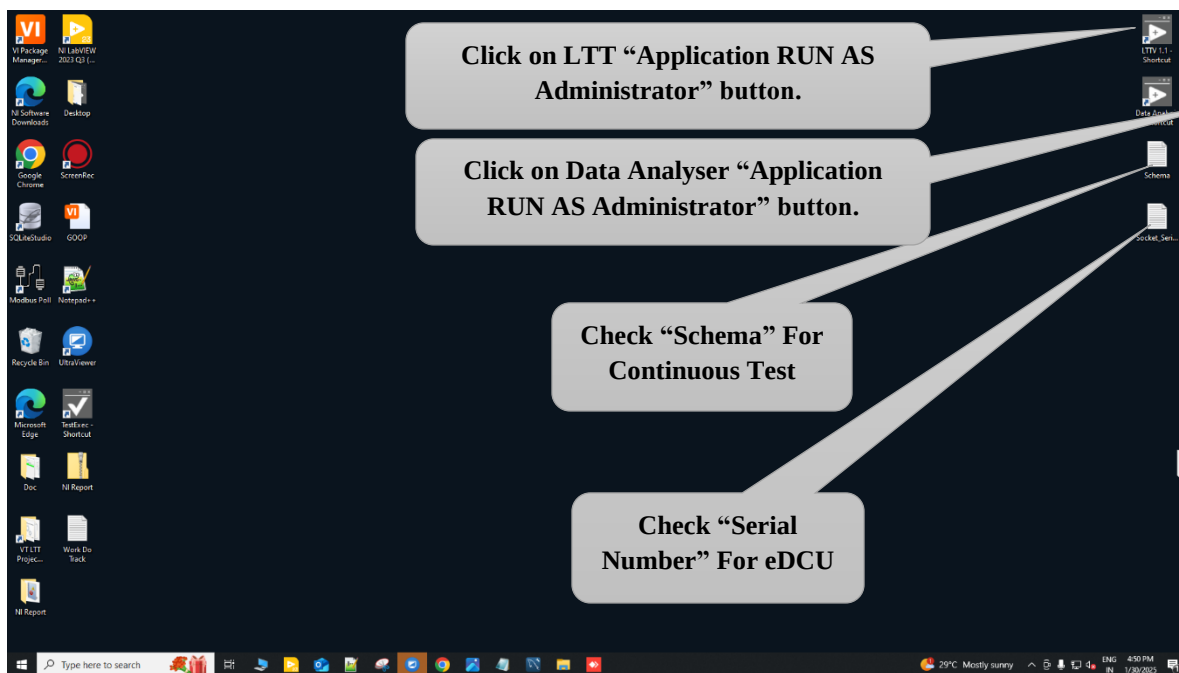


Figure 1: Desktop Panel

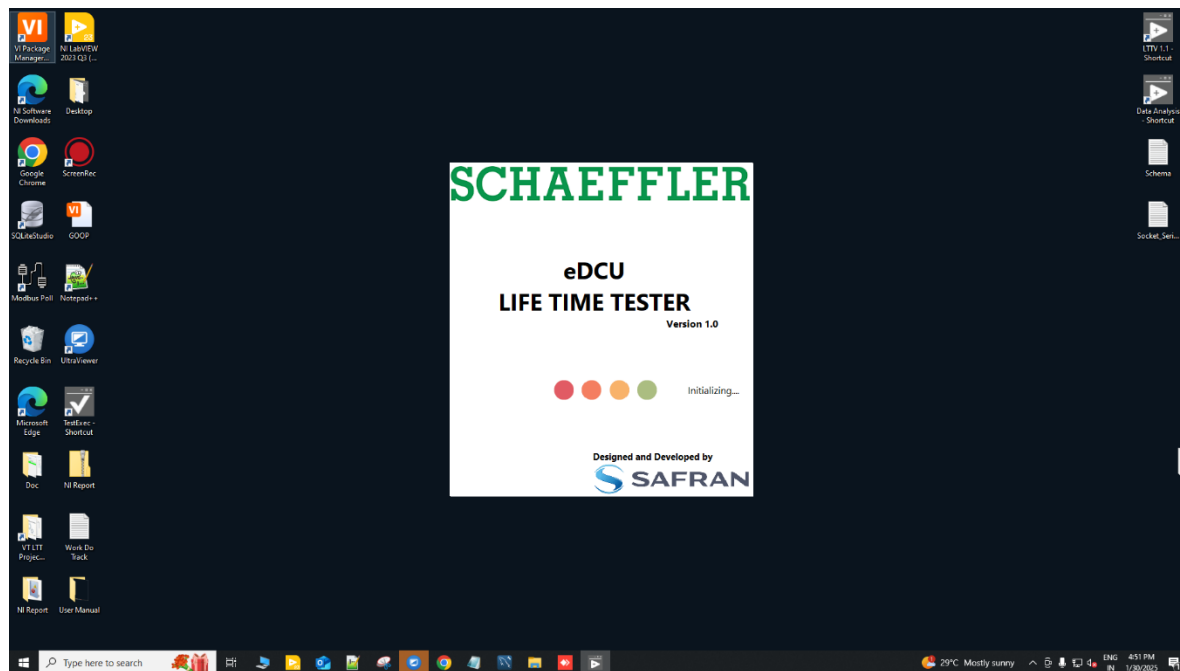


Figure 2: Loading Application

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 5

1.2 Main Screen::

Clicking the “Login” Button from the main screen below the user screen will appear.
Click “OK”.

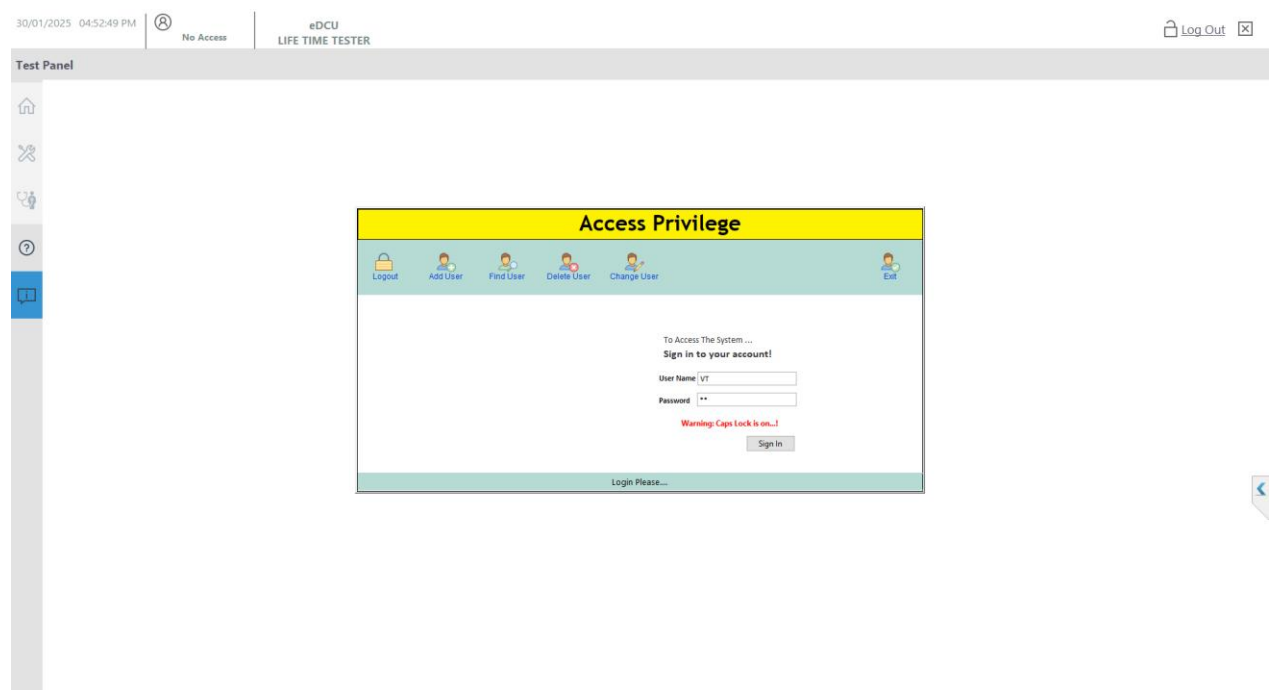


Figure 3: Main User Screen

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 6

1.3 Main Login:

Enter the “User Name” and “Password” Button from the Login screen below the user screen will appear. Click “Sign In”.

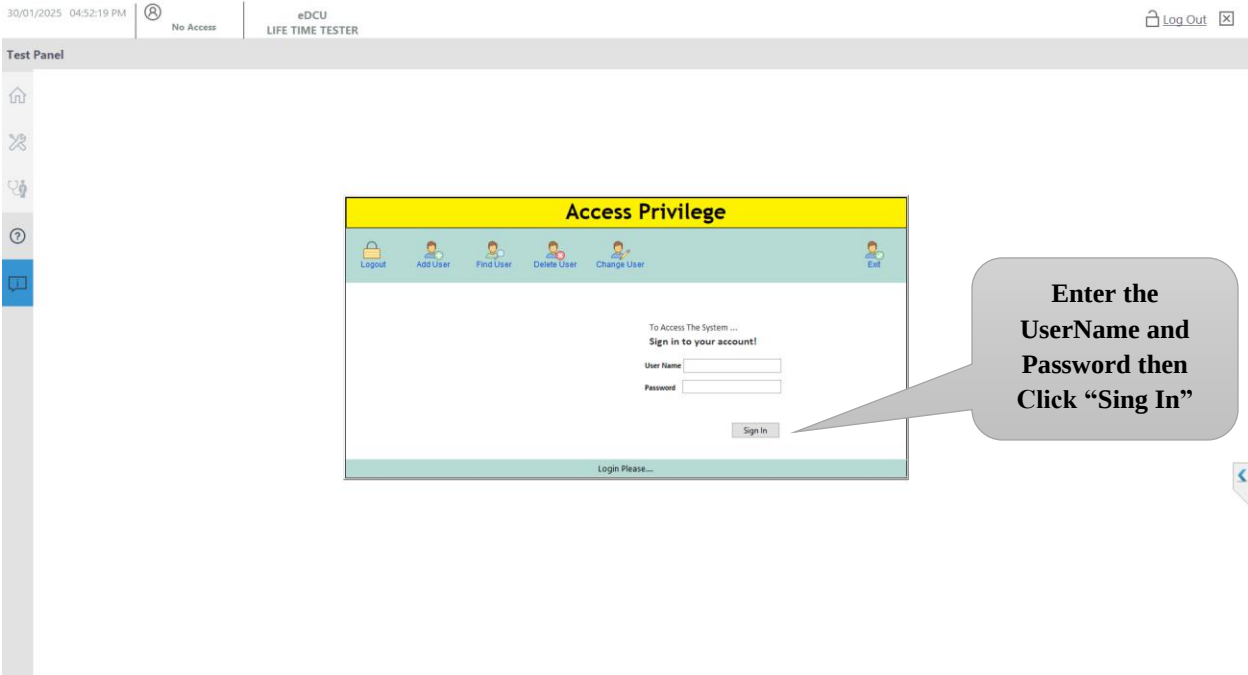


Figure 4: Login User Screen

Username: VT

Password: VT

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 7

1.4 Main Login Exit

Clicking the “Login OK” Button from the main screen below the user screen will appear. Click “OK”.

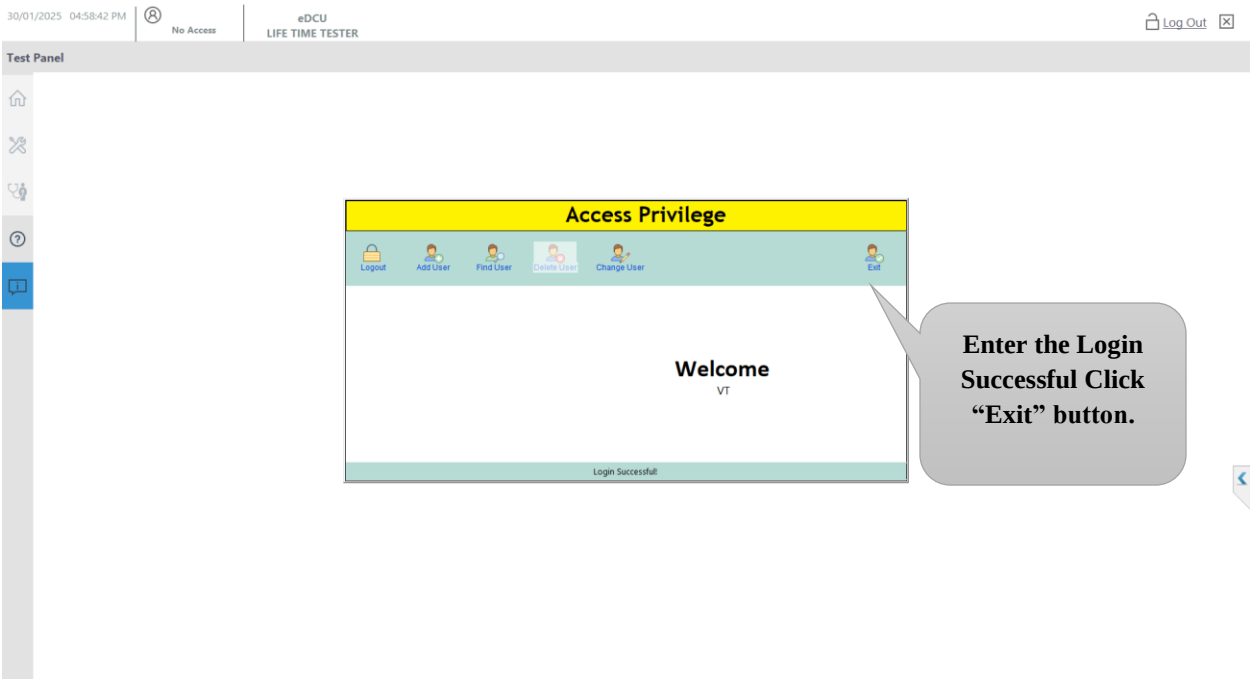


Figure 5: Login Successful

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 8

1.5 Auto Mode:

Clicking the “Login Successful” Button from the Click ”Auto mode” screen below the user screen will appear. “Auto Mode GUI”.

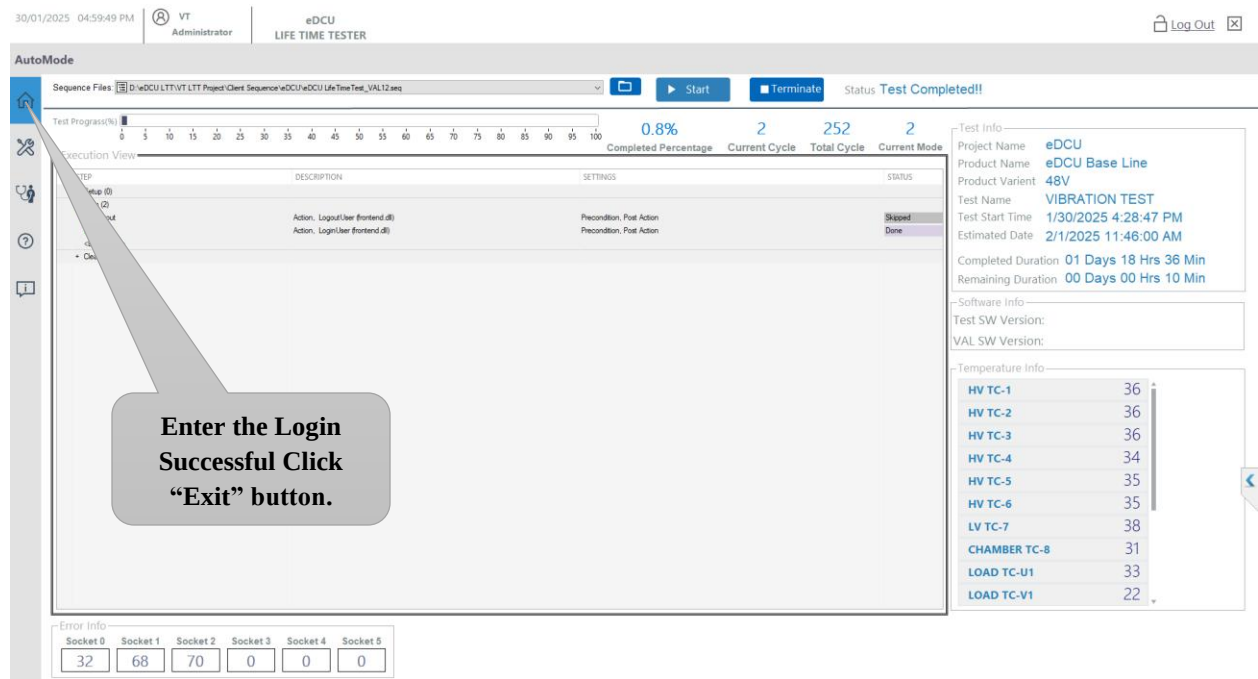


Figure 6: Auto Mode GUI Screen

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 9

1.6 Test stand Login:

The “Auto Mode Successful” Button from the Click ”TestStand Login” screen below the user screen will appear. “Login”.

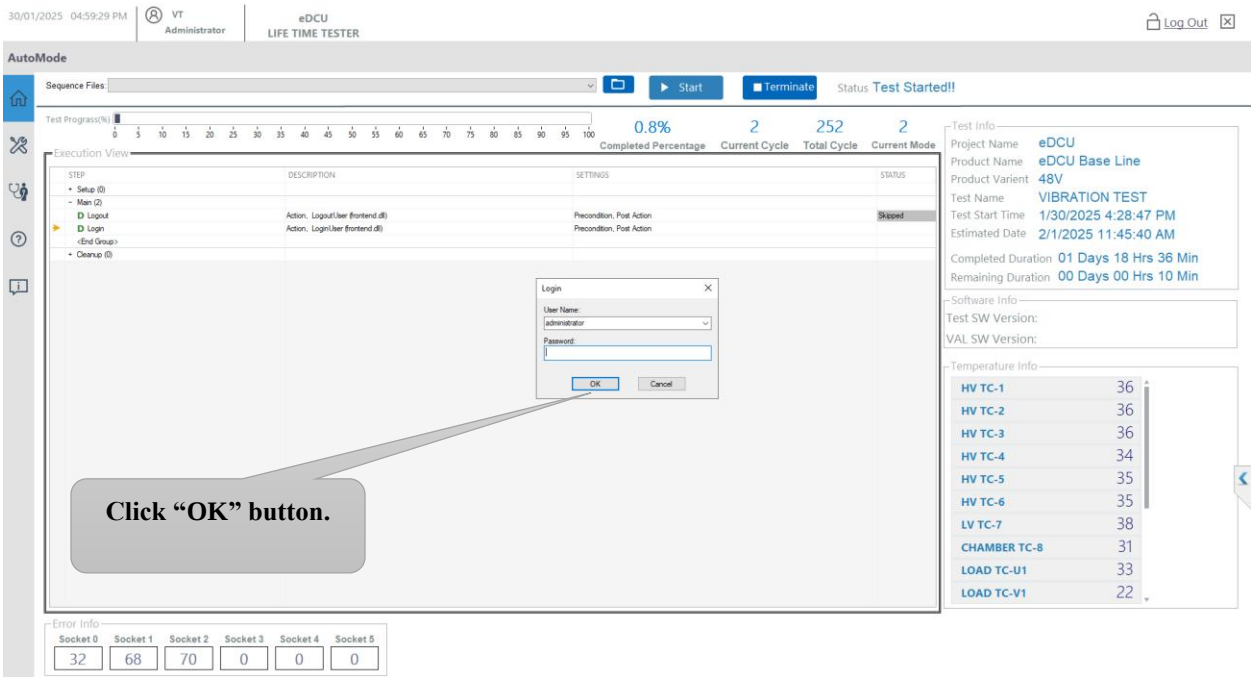


Figure 7: Test stand Login User Screen

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 10

1.7 Sequence Selection:

Press the Open Sequence Files then Select the “eDCU LifeTimeTest.seq” Sequence from the path. screen below the user screen will appear. Click “OPEN”

Path: D:\eDCU LTT\VT LTT Project\Client Sequence\eDCU

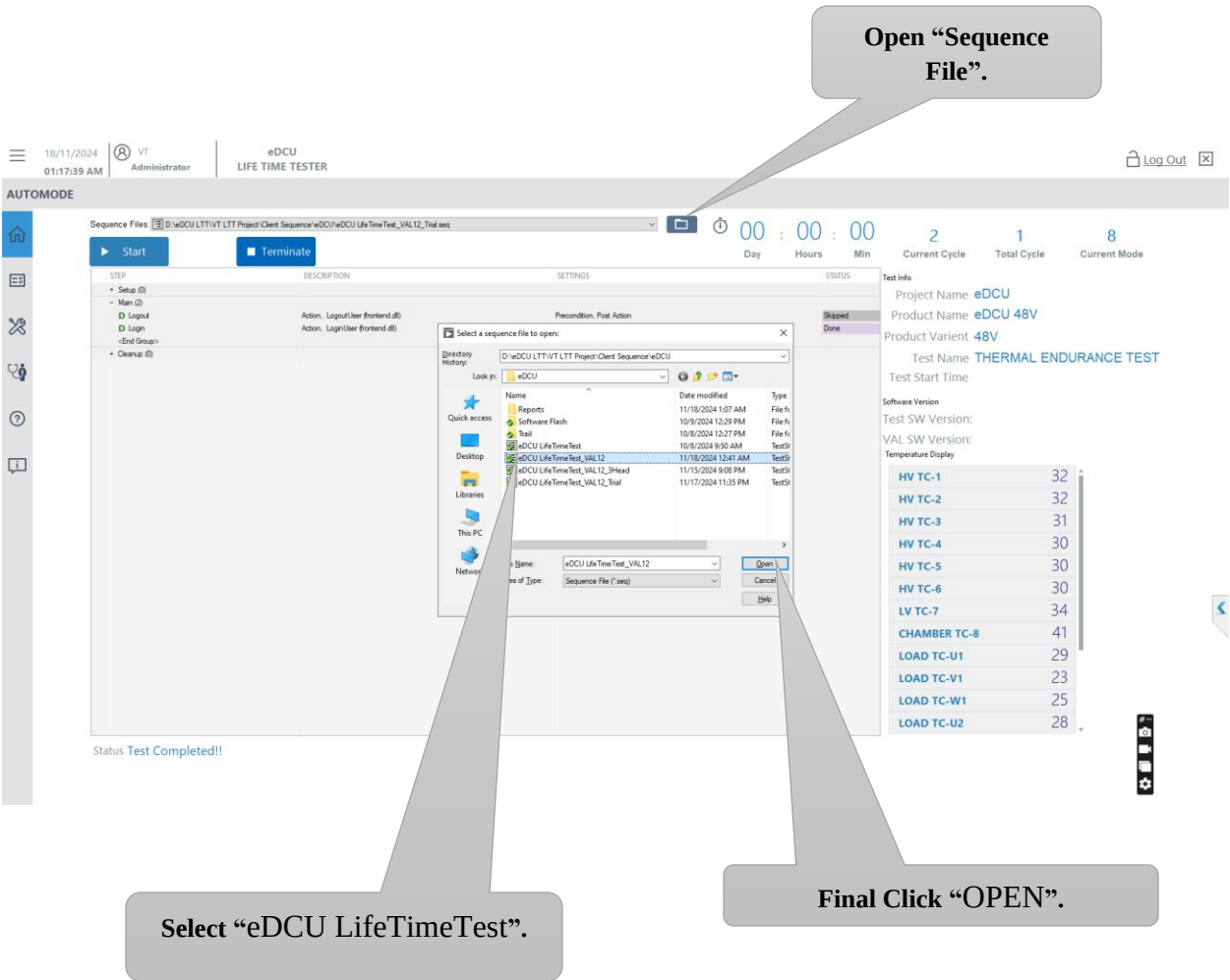


Figure 8: Sequence Selection

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 11

1.8 Test Windows:

Press the “Start” Sequence from the path. screen below the user screen will appear.

Path: D:\eDCU LTT\VT LTT Project\Client Sequence\eDCU\ eDCU LifeTimeTest.seq

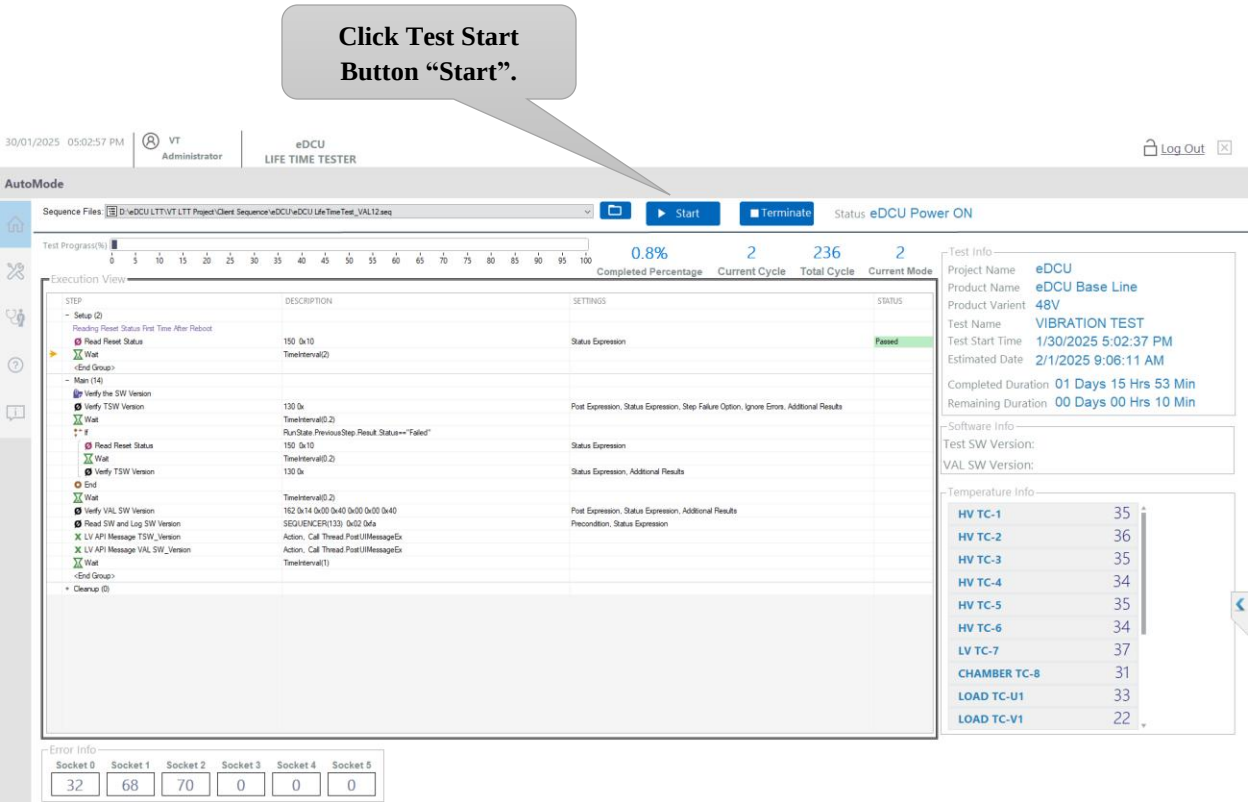


Figure 9: Start Test

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 12

1.9 Socket Entry Windows:

Press the “OK” IF “6”. The screen below the user screen will appear.

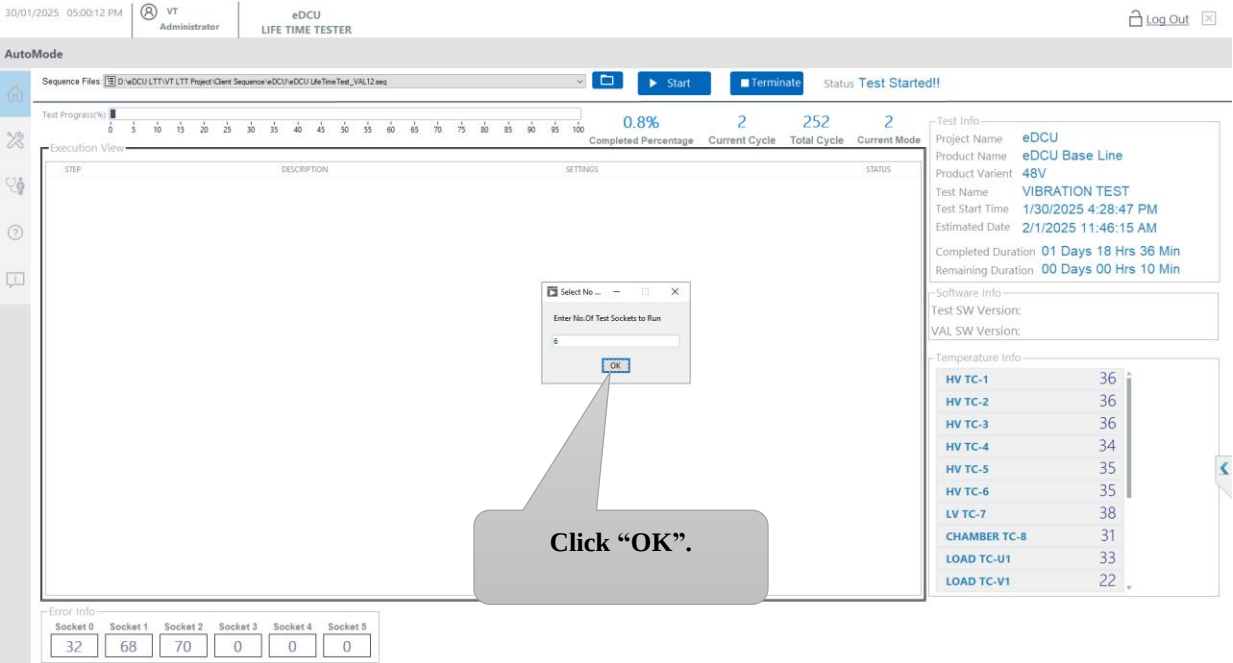


Figure 10: Socket Entry

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 13

1.10 UUT Info Entry Windows:

Enter the “Batch Number ” and “All Serial Number Enter”. The screen below the user screen will appear.

Batch No “Any Number”.

Enter all Serial No “Any Number”.

30/01/2025 05:00:54 PM VT Administrator eDCU LIFE TIME TESTER Log Out

AutoMode Sequence Files D:\eDCU LTT\VT LTT Project\Clert Sequence\edcuLifeTimeTest_VAL12.seq Start Terminate Status Select the Socket and Entry the Serial Number

Test Progress(%) 0.8% Completed Percentage 2 Current Cycle 252 Total Cycle 2 Current Mode 2

Execution View STEP DESCRIPTION

UUT Information

Batch Serial Number

Test Socket	UUT Serial Number	Status Message
0		Enter the next UUT serial number. ☐ Disable Test Socket
1		Enter the next UUT serial number. ☐ Disable Test Socket
2		Enter the next UUT serial number. ☐ Disable Test Socket

Go Stop

Test Info

Project Name eDCU
Product Name eDCU Base Line
Product Variant 48V
Test Name VIBRATION TEST
Test Start Time 1/30/2025 4:28:47 PM
Estimated Date 2/1/2025 11:47:02 AM
Completed Duration 01 Days 18 Hrs 36 Min
Remaining Duration 00 Days 00 Hrs 10 Min

Software Info

Test SW Version:
VAL SW Version:

Temperature Info

HV TC-1	36
HV TC-2	36
HV TC-3	36
HV TC-4	34
HV TC-5	35
HV TC-6	35
LV TC-7	38
CHAMBER TC-8	31
LOAD TC-U1	33
LOAD TC-V1	22

Error Info

Socket 0	Socket 1	Socket 2	Socket 3	Socket 4	Socket 5
32	68	70	0	0	0

Figure 11: UUT Information

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 14

1.11 UUT Info Entry Scroll Windows:

Enter the “Batch Number” and “All Serial Number Enter” and Proceed to “GO” The screen below the user screen will appear.

The screenshot displays the AUTOMODE interface of the eDCU Life Time Tester. A modal window titled "UUT Information" is open, allowing for the entry of test parameters. The window includes a "Batch Serial Number" field and a table for "Test Socket" and "UUT Serial Number". The table has five rows, indexed 0 to 4. Each row contains a "Test Socket" column, a "UUT Serial Number" input field, and a "Status Message" column. Below the table, there are "Go" and "Stop" buttons. A status bar at the bottom of the window reads "Status: Select the Socket and Entry the Serial Number". To the right of the modal, a "Test Info" panel displays project details: Project Name (eDCU), Product Name (eDCU 48V), Product Variant (48V), Test Name (THERMAL ENDURANCE TEST), Test Start Time, Software Version, Test SW Version, VAL SW Version, and Temperature Display. A list of temperature points is shown on the right, including HV TC-1 through LV TC-7, CHAMBER TC-8, and various LOAD TCs, each with a corresponding value.

Test Socket	UUT Serial Number	Status Message
0	2024052902017	Enter the next UUT serial number.
1	2024052902007	Enter the next UUT serial number.
2	2024052902043	Enter the next UUT serial number.
3	2024052902035	Enter the next UUT serial number.
4	2024052902038	Enter the next UUT serial number.

After all Entry only
Click “GO”.

Figure 12: GO Test

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 15

1.12 Test Selection Windows:

Click the “Continuous Test”. The screen below the user screen will appear.

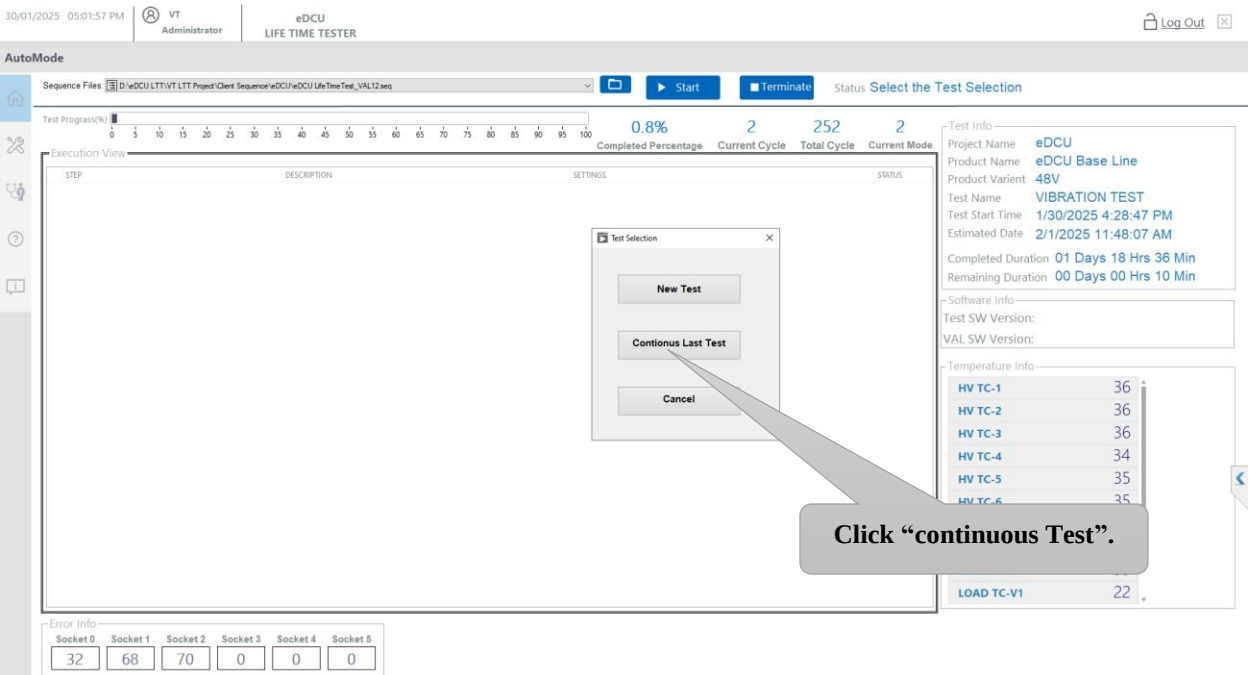


Figure 13: Continuous Test Selection

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 16

1.13 Schema File Location:

Select the schema “File mention the name”. The screen below the user screen will appear.

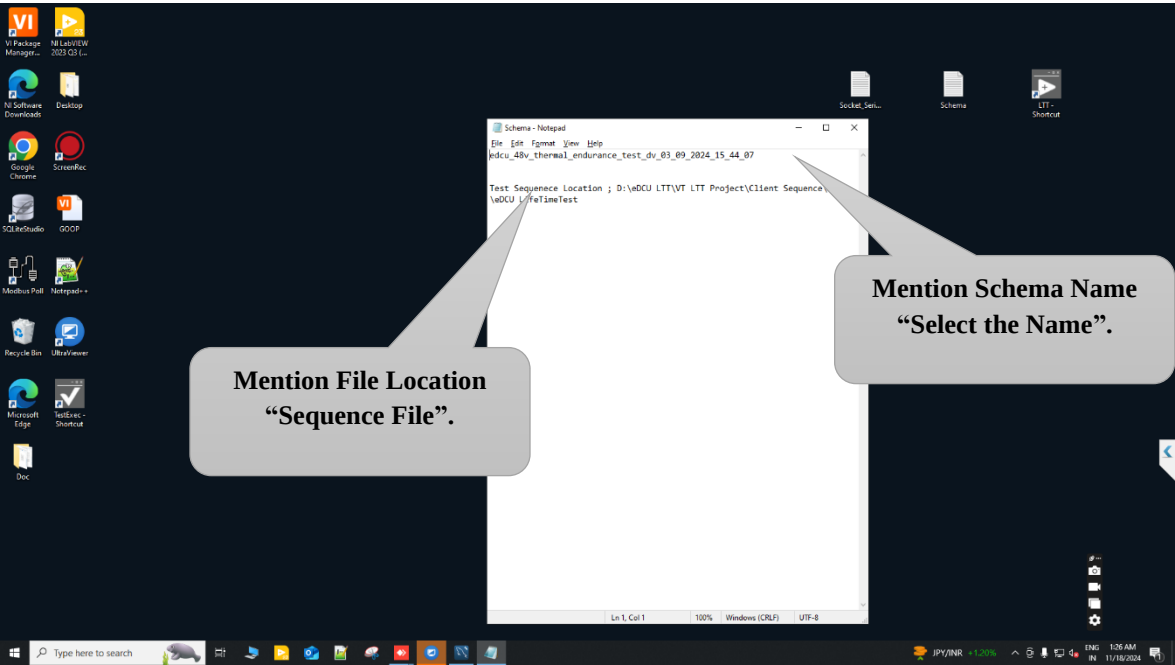


Figure 14: File Location

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 17

1.14 Schema Selection Windows:

Select the schema “File mention the name Schema Select Dropdown bar”. The screen below the user screen will appear.

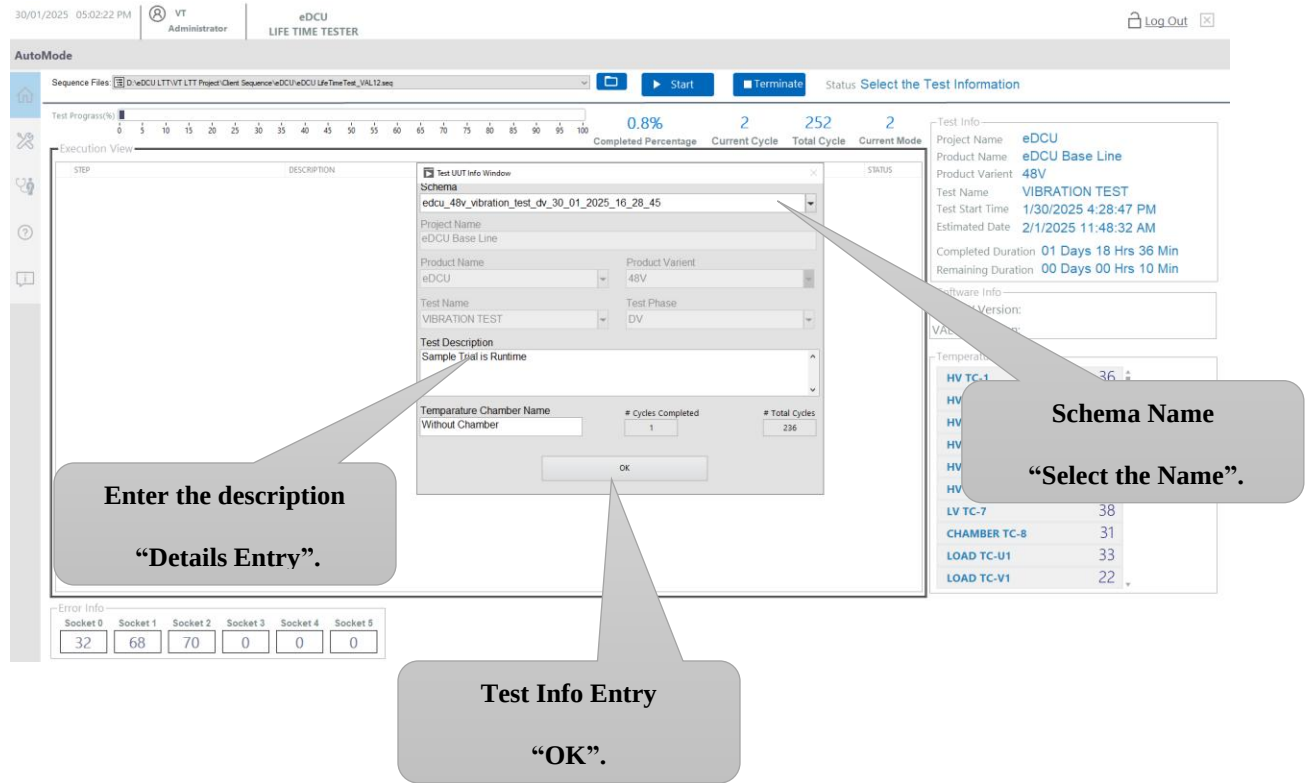


Figure 15: Test Info Schema Selection

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 18

1.15 Start the Chamber and Test Start Windows:

Select the schema “File mention the name”. Start the Chamber and when the chamber temperature targets is 50 deg set then click the OK Button. The screen below the user screen will appear.

This is applicable for thermal endurance test.

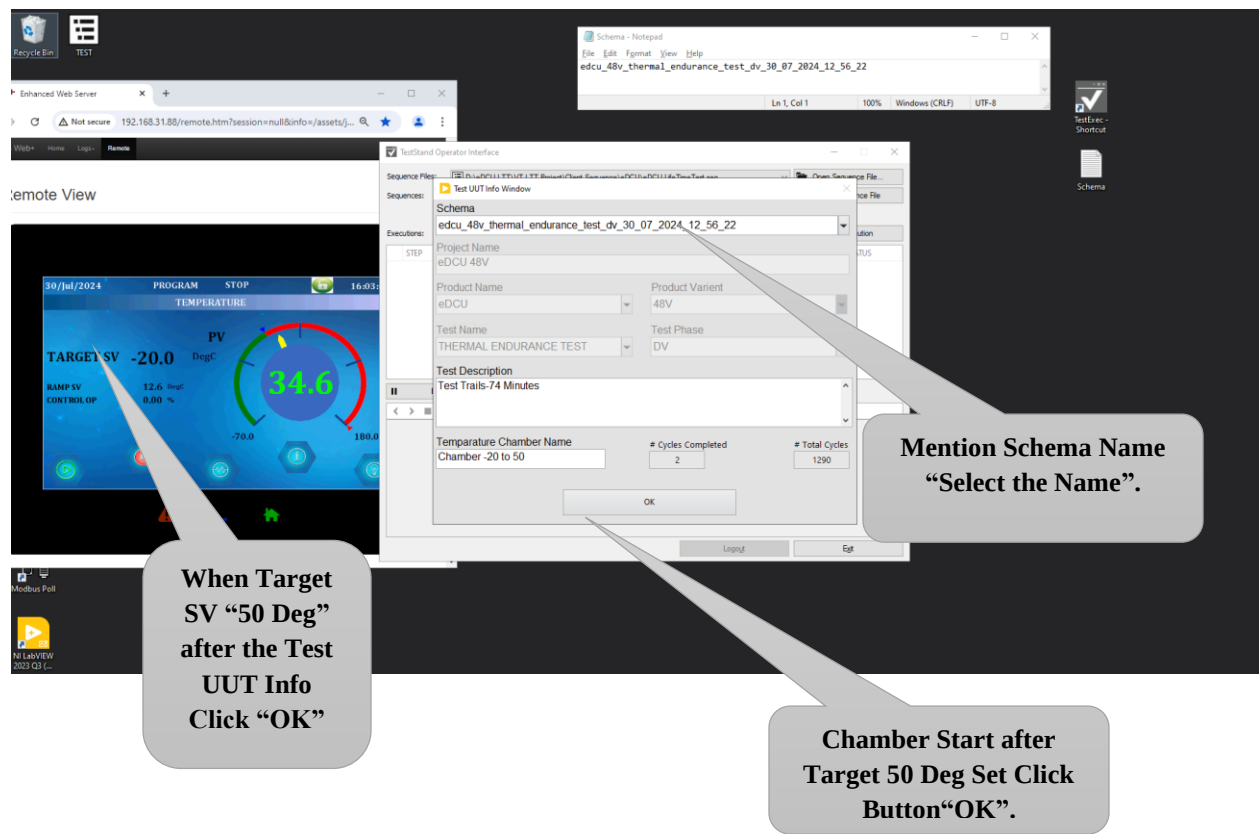
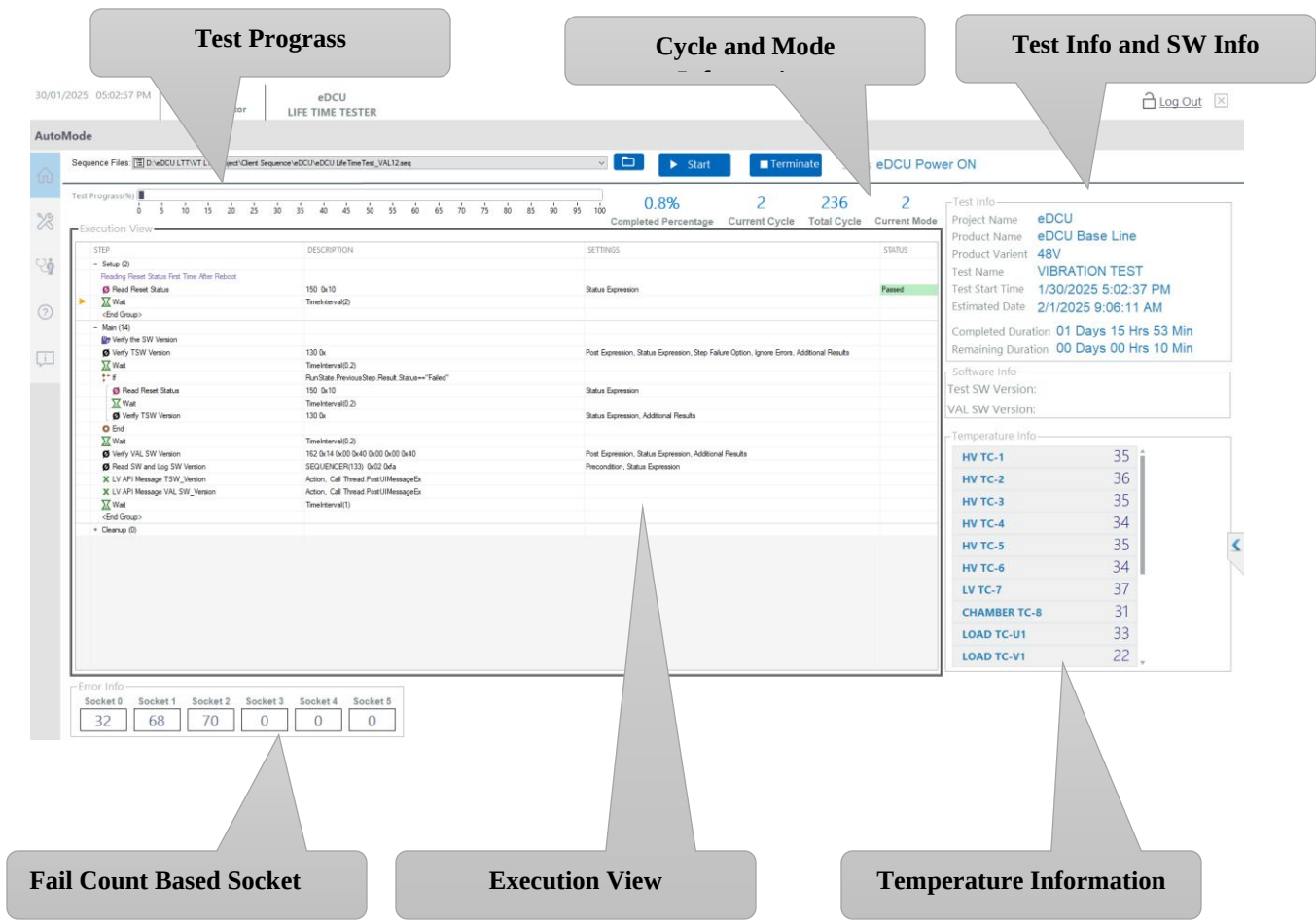


Figure 16: Stat and Click Ok

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 19

1.16 Running Windows of Monitor:

Check the Display the Status Update “Running Display”. Any message showing please inform the concern person. screen below the user screen will appear.



Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 20

1.17 Terminate Test Procedure:

Terminate the test Double Click Terminate the button and Terminate the test. Any message showing please inform the concern person. screen below the user screen will appear.

30/01/2025 05:04:06 PM VT Administrator eDCU LIFE TIME TESTER Log Out

AutoMode

Sequence Files D:\eDCU\LTT\VT\LTT Project Client Sequence\edcu\LifeTimeTest_VAL12.seq Start Terminate Status eDCU Power ON

Test Progress 0.8% Completed Percentage 236 Current Cycle 12 Current Mode

STEP	DESCRIPTION	SETTINGS	STATUS
Setup (0)			
Main (0)			
Set Given Mission Profile Mode	133 0x05 0x03 0x0e	Precondition, Pre-Expression, Post-Expression, Status Expression	Failed
If eDCU Set Profile Mode Fails it was call fail Seq			
Call FAIL Sequence	Call FAIL Sequence in <Current File>		Skipped
Updating the Test Mode LVQ and Station Globals	Action: Result FG V in TimeInterval(2)		Done
Ramp Up Time -2 Secs			
End Group			
Cleanup (0)			
High Voltage Power Supply	Measure		
Low Voltage Power Supply	Measure		
Calculate Time Lapsed Since Mode Start	FileGlobal TimeLapsedSinceModeStart - Second(0)		
Start Mode			
End Group			

Are You Sure Terminate? OK Cancel

Test Info

Project Name eDCU

Product Name eDCU Base Line

Product Variant 48V

Test Name VIBRATION TEST

Test Start Time 1/30/2025 5:02:37 PM

Estimated Date 2/1/2025 9:07:16 AM

Completed Duration 01 Days 15 Hrs 5

Remaining Duration 00 Days 00 Hrs 1

Software Info

Test SW Version: STZE0_OT0_030_

VAL SW Version: FS_STZE0_SEQ_

Temperature Info

HV TC-1 35

HV TC-2 36

CHAMBER TC-8 31

LOAD TC-U1 33

LOAD TC-V1 22

Socket 1 73 Socket 2 0 Socket 3 0 Socket 4 0 Socket 5 0

Double Click the "Terminate All". Please Inform and Need to Terminate Test

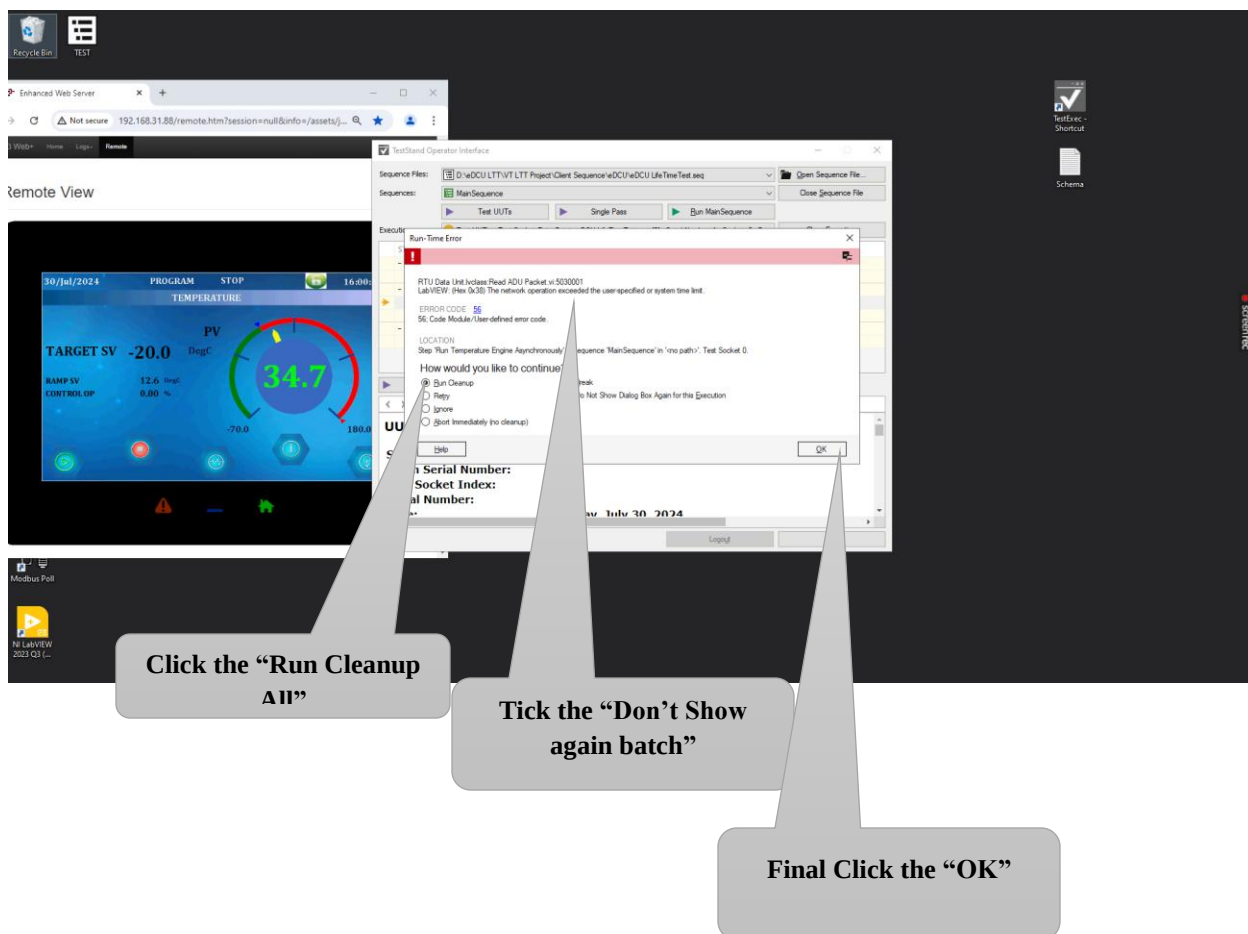
Click the "YES or NO"

Wait 10 Secs then Final Click the "EXIT"

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 21

1.18 Error Pop-UP Test Procedure:

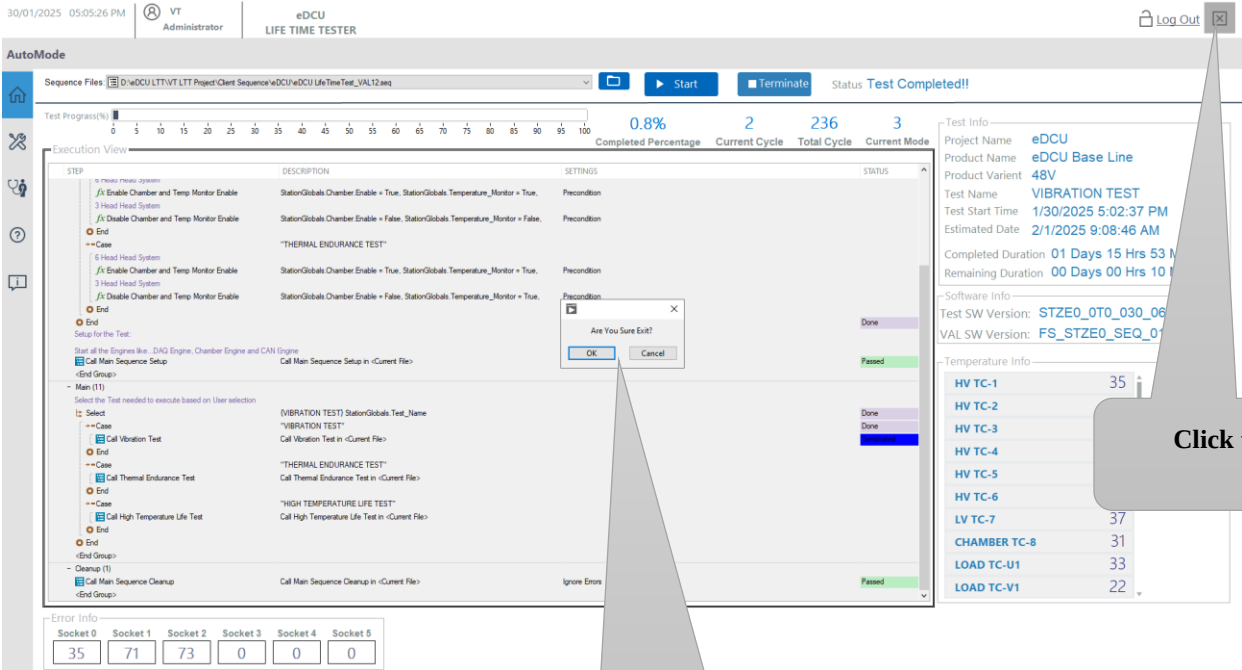
If any error popup like come so Select the Run Cleanup and tick the Don't show again batch execution then click the OK. screen below the user screen will appear.



Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 22

1.19 Exit Application:

If Exit the application Click the Exit button flowing the procedure. screen below the user screen will appear.



Click the "Exit"

Final Click the "OK or Cancel"

Project Name: eDCU Life Time Tester	Document ID: 23-10-051-VTI-CCAIC-UM	Version:2.0
	PP Form No: R/PP-03/25	Version Date: 18-11-2024
Type of document: Operation User Manual		Page 23